

The research area that interests me the most is condensed matter physics. I am particularly passionate about electronic devices and 2D materials. During my undergraduate and master's studies, I worked with 2D materials, and the lab I was part of also conducted research on organic photovoltaics (OPVs) and gas sensors. Additionally, I have also worked with lithium-ion batteries at the SENAI Institute for Innovation in Brazil.

While I have a deep interest in these areas, I am also interested in research involving other electronic devices, such as light-emitting diodes (LEDs) and transistors.

I am particularly intrigued by projects that explore next-generation photovoltaics and energy storage solutions. In this regard, the following projects resonate strongly with my interests and expertise:

- "Towards Pb-free perovskites for solar energy applications"(Dr. Julie Euvrad): Developing sustainable alternatives to Pb-based perovskites is an exciting theme for environmentally friendly energy solutions.
- "Interface engineering for high performance energy storage applications"(Prof. Ji-Seon Kim): My background in batteries aligns with the premise of this project.
- Both "Single-Component Organic Photovoltaics and Photodetectors"(Prof. Ji-Seon Kim) and "2D/3D Perovskites for Efficient Solar Cells"(Prof. Ji-Seon Kim) align closely with my background in characterizing materials for photovoltaic applications.
- "Device physics of single-component organic solar cells" (Prof. Jenny Nelson): Understanding the fundamental physics of organic materials offers a pathway to designing more efficient solar devices.

The research environment at Imperial College London is state-of-the-art, and its focus on sustainable energy perfectly aligns with my aspirations. I look forward to contributing to these research areas by leveraging my expertise and learning from leading scientists. My ultimate goal is to advance efficient, sustainable, and scalable energy technologies.